Algebraic Topology - Seifert-Van Kampen Theorem

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1 Free Product

Let G and H be groups. The free product of G and H, denoted by $G * H$, is defined as the set of reduced words.
A word is an expression of the form $g_1 g_2 \cdots g_n$, where $g_i \in G$ or $g_i \in H$ for all $i = 1, 2, \ldots, n$.
We can reduce a word by repeatedly applying the following two operations:

(i) If adjacent letters g_i and g_{i+1} belong to the same group, we replace them with their product $g_i g_{i+1}$, thereby reducing an n-letter word to an $(n-1)$-letter word.

(ii) If a letter g_j is the identity element e, we remove it, reducing an n-letter word to an $(n-1)$-letter word.

The multiplication of two words is defined by juxtaposition:

$$(g_1 g_2 \cdots g_n) \cdot (g_{n+1} g_{n+2} \cdots g_m) = g_1 g_2 \cdots g_n g_{n+1} \cdots g_m.$$

The identity element e is the empty word, and the inverse of $g_1 g_2 \cdots g_n$ is given by $g_n^{-1} g_{n-1}^{-1} \cdots g_1^{-1}$.
The set of all fully reduced words under this operation forms the free product $G * H$.
Similarly, we can define the free product of a family of groups $\{G_\alpha\}$ as

$$*_\alpha G_\alpha = \{g_1 g_2 \cdots g_n : \text{for each } i, \text{ there exists } \alpha \text{ such that } g_i \in G_\alpha\}.$$

We now verify that multiplication is associative in $*_\alpha G_\alpha$. Let W denote the set of all reduced words in $*_\alpha G_\alpha$. For each $g \in G_\alpha$, define a map $L_g : W \to W$ by $L_g(g_1 g_2 \cdots g_n) = g g_1 g_2 \cdots g_n$. For $g, g' \in G_\alpha$ and $g_1 g_2 \cdots g_n \in W$, we have

$$L_{g'}(L_g(g_1 g_2 \cdots g_n)) = g'(g(g_1 g_2 \cdots g_n)) = (g'g)(g_1 g_2 \cdots g_n) = L_{g'g}(g_1 g_2 \cdots g_n).$$

Thus $L_{g'g} = L_{g'} \circ L_g$, and it follows that L_g is invertible. We define a homomorphism $L : W \to \mathrm{Perm}(W)$ by $L(g_1 g_2 \cdots g_n) = L_{g_1} \circ L_{g_2} \circ \cdots \circ L_{g_n}$, where $\mathrm{Perm}(W)$ is the group of all permutations on W.
Suppose $u, v \in W$ satisfy $L(u) = L(v)$. Then $u = L(u)(e) = L(v)(e) = v$, demonstrating that L is injective.
Let $u = g_1 g_2 \cdots g_n$, $v = h_1 h_2 \cdots h_m$, and $w \in W$ such that $uv \in W$. Then

$$L(u)(L(v)(w)) = L(u)(h_1 h_2 \cdots h_m w) = g_1 g_2 \cdots g_n h_1 h_2 \cdots h_m w = L(uv)(w),$$

which implies $L(u) \circ L(v) = L(uv)$. Because function composition is inherently associative in $\mathrm{Perm}(W)$, for any $u, v, w \in W$, we have:

$$L(u(vw)) = L(u) \circ L(vw) = L(u) \circ (L(v) \circ L(w)) = (L(u) \circ L(v)) \circ L(w) = L(uv) \circ L(w) = L((uv)w).$$

Since L is injective, it follows that $(uv)w = u(vw)$. Therefore, multiplication in $*_\alpha G_\alpha$ is associative.

2 Lebesgue Number Theorem

Lemma 2.1 (Lebesgue Number Theorem)**.** *Let $\mathcal{A}$ be an open covering of a compact metric space (X, d). There exists a positive real number δ, called a Lebesgue number, such that every subset $C \subset X$ with diameter less than δ satisfies $C \subset A$ for some $A \in \mathcal{A}$.*

Proof. Let $\mathcal{A}$ be an open covering of X. For each $x \in X$, there exists some open set $A_x \in \mathcal{A}$ containing x. Since A_x is open, there exists a $\delta_x > 0$ such that the open ball $B_d(x, \delta_x) \subset A_x$. The family of open balls $\{B_d(x, \delta_x/2) : x \in X\}$ forms an open covering of X. Since X is compact, there exists a finite subcovering $\{B_d(x_i, \delta_{x_i}/2) : i = 1, \ldots, n\}$. We claim that $\delta = \min_{1 \le i \le n}\{\delta_{x_i}/2\}$ is a Lebesgue number for $\mathcal{A}$.
Let C be a subset of X with diameter less than δ. Choose any $x \in C$. Because the finite subcovering covers X, we must have $x \in B_d(x_i, \delta_{x_i}/2)$ for some i. For any other element $y \in C$, the triangle inequality yields:

$$d(y, x_i) \le d(y, x) + d(x, x_i) < \delta + \frac{\delta_{x_i}}{2} \le \frac{\delta_{x_i}}{2} + \frac{\delta_{x_i}}{2} = \delta_{x_i}.$$

Thus, $C \subset B_d(x_i, \delta_{x_i}) \subset A_{x_i} \in \mathcal{A}$, completing the proof. □

3 Seifert-Van Kampen Theorem

Theorem 1 (Seifert-Van Kampen)**.** *Let X be the union of path-connected open sets A_α, each containing the basepoint $x_0 \in X$. If each intersection $A_\alpha \cap A_\beta$ is path-connected, then*

the homomorphism $\Phi : *_\alpha \pi_1(A_\alpha, x_0) \to \pi_1(X, x_0)$ *induced by the inclusions* $j_\alpha : A_\alpha \hookrightarrow X$ *is surjective.*
If, in addition, each triple intersection $A_\alpha \cap A_\beta \cap A_\gamma$ *is path-connected, then the kernel of* Φ *is the normal subgroup* N *generated by all elements of the form* $i_{\alpha\beta}(w) i_{\beta\alpha}(w)^{-1}$ *for* $w \in \pi_1(A_\alpha \cap A_\beta, x_0)$*, where* $i_{\alpha\beta} : \pi_1(A_\alpha \cap A_\beta, x_0) \to \pi_1(A_\alpha, x_0)$ *is the homomorphism induced by the inclusion* $A_\alpha \cap A_\beta \hookrightarrow A_\alpha$*. Hence,* Φ *induces an isomorphism*

$$\Phi_* : (*_\alpha \pi_1(A_\alpha, x_0))/N \xrightarrow{\sim} \pi_1(X, x_0).$$

Proof. The proof of surjectivity is standard and is omitted here. We focus on showing that $\ker(\Phi) = N$.
First, note that $j_\alpha \circ i_{\alpha\beta} = j_\beta \circ i_{\beta\alpha}$ since both compositions are induced by the inclusion $A_\alpha \cap A_\beta \hookrightarrow X$. For any $w \in \pi_1(A_\alpha \cap A_\beta, x_0)$, we have:

$$\begin{aligned} \Phi(i_{\alpha\beta}(w) i_{\beta\alpha}(w)^{-1}) &= \Phi(i_{\alpha\beta}(w)) \Phi(i_{\beta\alpha}(w)^{-1}) \\ &= (j_\alpha i_{\alpha\beta}(w))(j_\beta i_{\beta\alpha}(w))^{-1} \\ &= (j_\alpha i_{\alpha\beta}(w))(j_\alpha i_{\alpha\beta}(w))^{-1} = e. \end{aligned}$$

Therefore, $N \subseteq \ker(\Phi)$, and Φ factors through the quotient to induce a homomorphism Φ_*. We now want to prove that Φ_* is injective. (If Φ_* is injective, then for any $g \in \ker(\Phi)$, we have $\Phi_*(gN) = \Phi(g) = e = \Phi(n)$ for some $n \in N$. This implies $gN = nN = N$, and thus $g \in N$, yielding $\ker(\Phi) \subseteq N$.)
To prove injectivity, we define two factorizations in $*_\alpha \pi_1(A_\alpha, x_0)$ to be equivalent if one can be transformed into the other via a finite sequence of the following operations:

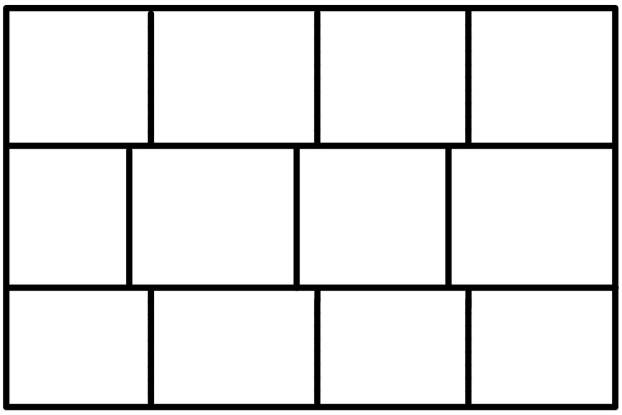
(i) If $[f_i], [f_{i+1}] \in \pi_1(A_\alpha, x_0)$, we may combine them into a single element $[f_i][f_{i+1}] = [f_i \cdot f_{i+1}]$.

(ii) If $[f_i] \in \pi_1(A_\alpha, x_0)$ is the image of an element in $\pi_1(A_\alpha \cap A_\beta, x_0)$, we may view it as an element $[f_i] \in \pi_1(A_\beta, x_0)$.

Notice that the equivalence class in $(*_\alpha \pi_1(A_\alpha, x_0))/N$ remains invariant under these operations.
Suppose $[f_1][f_2] \cdots [f_l]$ and $[f'_1][f'_2] \cdots [f'_k]$ are two words in $*_\alpha \pi_1(A_\alpha, x_0)$ that map to the same element in $\pi_1(X, x_0)$. That is, their compositions are path-homotopic in X. Let $F : I \times I \to X$ be a path homotopy such that

$$F(t, 0) = (f_1 \cdot f_2 \cdots f_l)(t), \qquad F(t, 1) = (f'_1 \cdot f'_2 \cdots f'_k)(t).$$

Since $I \times I$ is a compact metric space, we may apply the Lebesgue Number Theorem. There exist partitions $0 = s_0 < s_1 < \cdots < s_m = 1$ and $0 = t_0 < t_1 < \cdots < t_n = 1$ such that each sub-rectangle $R_{i,j} = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ satisfies $F(R_{i,j}) \subset A_\alpha$ for some α. We assign each $R_{i,j}$ an open set $A_{i,j}$ containing its image.
Furthermore, we can perturb the vertices of this grid so that each vertex is the intersection of at most three rectangles. We order the rectangles from left to right, and from bottom to top, indexing them as $R'_1, \ldots, R'_{nm}$.



Let γ_i be the word representing the path in $I \times I$ separating the first i rectangles from the remaining ones, evaluated under F. Note that $\gamma_0 = F|_{I\times\{0\}} = (f_1 \cdot f_2 \cdots f_l)(t)$ and $\gamma_{nm} = F|_{I\times\{1\}} = (f'_1 \cdot f'_2 \cdots f'_k)(t)$.
Since each vertex lies in at most three rectangles, its image falls in the intersection $A_\alpha \cap A_\beta \cap A_\gamma$ for some α, β, γ. We can connect each vertex v to the basepoint x_0 by a path g_v lying entirely within this intersection. If $F(v) = x_0$, we choose g_v to be the constant path.
We claim that γ_{i+1} can be obtained from γ_i by a finite sequence of operations (i) and (ii). The paths γ_i and γ_{i+1} differ only along the boundary of the rectangle R'_{i+1}. Because $F(R'_{i+1}) \subset A_\alpha$ for some α, we can use operation (ii) to view the relevant boundary segments of both γ_i and γ_{i+1} as loops in A_α. Since these segments are homotopic within A_α (via the interior of R'_{i+1}), operation (i) allows us to replace the segment in γ_i with the segment in γ_{i+1}.
Applying this argument inductively from γ_0 to γ_{nm} demonstrates that the initial and final words are equivalent under (i) and (ii), meaning they define the same element in the quotient group. Therefore, Φ_* is injective. By the First Isomorphism Theorem, we have

$$(*_\alpha \pi_1(A_\alpha, x_0))/N \cong \pi_1(X, x_0).$$

□

4 Applications

Proposition 4.1. *Let X be a topological space and A a subspace of X. If A is a deformation retract of X, then the homomorphism $i_* : \pi_1(A, x_0) \to \pi_1(X, x_0)$ induced by the inclusion $i : A \hookrightarrow X$ is an isomorphism.*

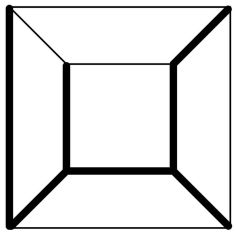
Proof. If $r : X \to A$ is a retraction, then $r \circ i = 1_A$, which implies $r_* \circ i_* = 1_{\pi_1(A,x_0)}$. This shows that i_* is injective.
Furthermore, if $r_t : X \to X$ is a deformation retraction of X onto A, we have $r_0 = 1_X$, $r_t|_A = 1_A$, and $r_1(X) \subset A$. For any loop $f : I \to X$ based at x_0, the composition $r_t \circ f$ provides a homotopy from f to a loop entirely contained in A. Thus, every loop in X is path-homotopic to a loop in A, demonstrating that i_* is surjective. □

Example 4.1 (Wedge Sums)**.** *Recall the definition of the wedge sum: Let* $\{X_\alpha\}$ *be a collection of spaces with basepoints* $x_\alpha \in X_\alpha$*. The wedge sum is defined as* $\bigvee_\alpha X_\alpha = \coprod_\alpha X_\alpha/\sim$*, where all basepoints* x_α *are identified to a single common basepoint.*
Assume that each X_α *is path-connected and that each* x_α *is a deformation retract of an open neighborhood* $U_\alpha \subset X_\alpha$*. We can set* $A_\alpha = X_\alpha \vee \bigvee_{\beta\neq\alpha} U_\beta$*. Notice that* X_α *is a deformation retract of* A_α*, and the intersection* $A_\alpha \cap A_\beta \cap A_\gamma = \bigvee_\alpha U_\alpha$*, which deformation retracts to a single point.*
By the Seifert-Van Kampen Theorem, the map $\Phi : *_\alpha \pi_1(A_\alpha, x_\alpha) \to \pi_1(\bigvee_\alpha X_\alpha, x_\alpha)$ *is an isomorphism. Because* X_α *is a deformation retract of* A_α*, we have* $\pi_1(A_\alpha, x_\alpha) \cong \pi_1(X_\alpha, x_\alpha)$*. Therefore,*

$$*_\alpha \pi_1(X_\alpha, x_\alpha) \cong *_\alpha \pi_1(A_\alpha, x_\alpha) \cong \pi_1\Big(\bigvee_\alpha X_\alpha, x_\alpha\Big).$$

In particular, $\pi_1(S^1 \vee S^1) \cong \pi_1(S^1) * \pi_1(S^1) \cong \mathbb{Z} * \mathbb{Z}$*. By induction,* $\pi_1(\bigvee_n S^1) \cong *_n \mathbb{Z}$*.*

Example 4.2 (Fundamental Group of a Graph)**.** *Let* X *be the graph consisting of the* 12 *edges of a cube. We can choose a maximal tree* $T \subset X$ *consisting of* 7 *edges;* T *is a contractible subgraph containing all vertices of* X*.*



For each of the 5 *edges* e_α *not in* T*, choose* A_α *to be an open neighborhood of* $T \cup e_\alpha$ *in* X *that deformation retracts onto* $T \cup e_\alpha$*. Note that* $\bigcup_\alpha A_\alpha = X$*, and any multiple intersection of the* A_α*'s deformation retracts onto the tree* T*, meaning it is path-connected and simply connected.*
By the Seifert-Van Kampen Theorem, $\pi_1(X) \cong *_\alpha \pi_1(A_\alpha)$*. Since each* A_α *deformation retracts onto a circle* S^1*, we obtain* $\pi_1(X) \cong *_5 \pi_1(S^1) \cong *_5 \mathbb{Z}$*. Thus,* $\pi_1(X)$ *is the free product of* 5 *copies of* $\mathbb{Z}$*.*